Requirement Analysis

Technology Stack (Architecture & Stack)

Date	20 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below along with the information in Tables 1 and 2.

Global Olympic Sports Performance Analysis – Architecture Flow

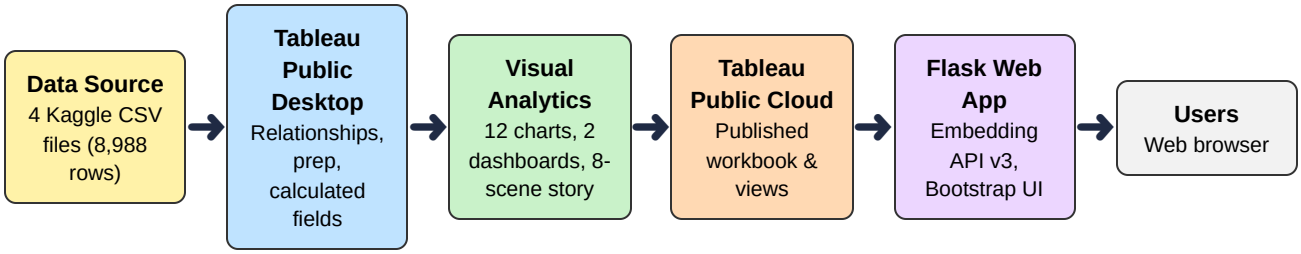


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How users view the dashboards and story	Flask web pages (HTML, CSS, Bootstrap 5), Tableau Public web view, Web browser
2.	Application Logic-1	Relates and prepares the 4 CSV files (types, nulls, discrete Year)	Tableau Public Desktop (Data Source)
3.	Application Logic-2	Builds visualizations, calculated fields, parameter, dashboards and story	Tableau Public Desktop
4.	Application Logic-3	Serves web pages and embeds published views	Python Flask, Tableau Embedding API v3
5.	Database	No traditional database; data comes from CSV files	Not Applicable for this project
6.	Cloud Database	Not Applicable for this project	Not Applicable for this project
7.	File Storage	Raw CSV files, Tableau workbook and Flask source code	Local File System (CSV, .twbx), GitHub
8.	External API-1	Embeds Tableau Public views into web pages	Tableau Embedding API v3 (JavaScript)
9.	External API-2	Not Applicable for this project	Not Applicable for this project
10.	Machine Learning Model	Not Applicable for this project	Not Applicable for this project

11.	Infrastructure (Server / Cloud)	Local development; dashboards hosted on Tableau Public; website run locally with Flask; code on GitHub	Local Machine, Tableau Public, Flask dev server, GitHub

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web framework and UI styling	Python Flask, Bootstrap 5
2.	Security Implementations	Only public, non-personal data; read-only public views; no credentials stored	Tableau Public, GitHub access control
3.	Scalable Architecture	Modular workbook (sheets → dashboards → story) and modular Flask routes	Tableau workbook, Flask templates
4.	Availability	Views available anytime through Tableau Public links	Tableau Public hosting
5.	Performance	Relationships instead of joins, Top-N filters and aggregated views for fast rendering	Tableau rendering optimization